TEXORA · LECTURE SHEET

Yarn Conditioning

Yarn leaves the ring frame dry, lively and stressed. Conditioning brings it to a controlled moisture content — setting its twist, restoring its strength, and making its weight, dyeing and weaving behaviour predictable.

TE-2201 · SPINNING TECHNOLOGY-II · TOPIC 6 · LEVEL 2 TERM 2

1. Why yarn is conditioned

Spinning rooms are run drier than the ideal for yarn, and friction in drafting and winding drives still more moisture out: a freshly spun cop typically sits well below the fibre's equilibrium regain. In that state the yarn is **lively** — its twist tries to untwist, forming snarls and kinks whenever the yarn goes slack — and its fibres are in internal stress. Conditioning is the deliberate return of moisture, evenly through the whole package, to a specified regain. Its purposes:

- **Twist setting:** moist fibres relax and re-form their hydrogen bonds in the twisted configuration, so the yarn keeps its twist without snarling — essential for crepe and voile yarns and for any yarn that will be handled slack.
- **Strength recovery:** cotton (and other cellulosic) yarn is stronger at its proper regain than bone dry; conditioning restores the strength that dry spinning cost.
- **Processing behaviour:** conditioned yarn runs with fewer breaks in winding, warping and weaving, develops less static electricity and less fly.
- **Uniform dyeing and finishing:** dye uptake follows moisture state; packages at one even regain dye to one even shade.
- **Honest weight:** yarn is sold by weight and count is a weight-per-length measure; conditioning brings the commercial weight to the agreed standard regain instead of whatever the spinning room left.

2. Moisture regain — the controlling quantity

$$\text{Moisture regain (\%)} = (\text{mass of water in the material} \div \text{oven-dry mass}) \times 100$$

Regain must not be confused with **moisture content**, which divides by the total (wet) mass. Textiles are traded and tested on regain, with internationally agreed **standard (commercial) regains** — for cotton 8.5 %. A material in air absorbs or releases moisture until it reaches the **equilibrium regain** for that

air's humidity and temperature — which is why testing is done in the **standard atmosphere**: 20 ± 2 °C and 65 ± 4 % relative humidity (ISO 139), after the sample has had time to condition to equilibrium in it.

> **WORKED EXAMPLE — CHECKING A REGAIN**
>
> A yarn sample weighs 103.0 g as received and 94.9 g after oven drying.
> Regain = (103.0 − 94.9) ÷ 94.9 × 100 = 8.1 ÷ 94.9 × 100 ≈ **8.5 %** — exactly the standard commercial regain for cotton. The same sample's moisture content would be 8.1 ÷ 103.0 ≈ 7.9 %: the two numbers differ, and specifications always mean regain unless they say otherwise.

3. How moisture changes yarn behaviour

PROPERTY	EFFECT OF RISING REGAIN (TO THE STANDARD RANGE)
Strength — cotton, linen, viscose	**Increases** (cellulosic fibres are stronger moist; cotton gains of the order of 10–20 % over the bone-dry state are typical). Wool and silk behave oppositely — they weaken slightly as they take up water.
Elongation at break	Increases: fibres extend more before failing, so yarn tolerates the tension peaks of weaving better.
Twist liveliness / snarling	Falls sharply once the twist is moisture-set; dry yarn is the snarling yarn.
Static and fly	Fall: a moisture film conducts charge away and binds fibre ends down.
Dye uptake	Becomes faster and, above all, more uniform when regain is uniform.
Weight and count	Rise with regain — an under-conditioned delivery is an under-weight delivery, and count tests mislead until samples are conditioned.

4. Conditioning methods

METHOD	HOW IT WORKS	CHARACTER
Conditioning room (atmospheric)	Packages rest on racks in a room held at high humidity (≈ 75–85 % RH) until they approach equilibrium — commonly a day or more for dense packages	Cheapest and gentlest, but slow, space-hungry, and the outside of a package conditions long before its core
Steam conditioning (autoclave)	Packages on trolleys enter a sealed vessel; saturated steam is admitted for a	Fast and effective twist setting; the cycle (temperature, time) is tuned per yarn —

	set cycle, heating and moistening the yarn together	excessive heat yellows fibres and damages wool and elastane
Vacuum steam conditioning	The vessel is first evacuated, *then* steam is admitted: with the air gone, steam penetrates the package instantly and uniformly	The modern standard — even core-to-surface regain in cycles of tens of minutes; best, most reproducible results
Radio-frequency / dielectric conditioning	The package passes through a high-frequency field that heats the water within the fibres selectively	Very even and rapid for some products; higher equipment cost, niche use

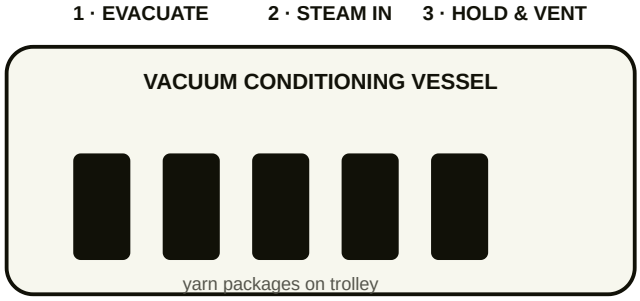


Figure 1 — Vacuum steam conditioning. Evacuating before steaming is what lets moisture reach the centre of a dense package; a plain steam or room cycle conditions the surface first and the core last.

5. Over- and under-conditioning

Under-conditioned yarn weaves badly: high end breaks, snarls and loops in the weft, static-driven fly, and an underweight delivery with count results that drift as the yarn later picks up moisture. **Over-conditioned yarn** is not safer: regain pushed well above standard invites **mildew** and bacterial damage in storage (cellulosics stain and tender), corrosion marks from metal tubes and containers, packages that stick and unwind badly, and dyeing that starts from an abnormal state. Conditioning targets the standard regain — with a small, deliberate tolerance — not "as much moisture as possible". After conditioning, packages should rest briefly so surface and core equalise before testing or dispatch, and conditioned stock is stored under cover, off the floor, and used in rotation.

Glossary

Regain — water mass as a percentage of oven-dry mass. **Moisture content** — water mass as a percentage of total mass (the smaller number). **Standard atmosphere** — 20 ± 2 °C, 65 ± 4 % RH, in which textile tests are performed. **Equilibrium regain** — the regain a material settles to in a given atmosphere. **Liveliness** — a yarn's tendency to snarl and kink when slack, from unset twist. **Twist**

setting — permanently relaxing a yarn's untwisting torque by heat and/or moisture. **Commercial (standard) regain** — the agreed regain (cotton 8.5 %) used in trade weight calculations.

Key points

• Conditioning returns yarn to a specified regain: it sets twist, recovers strength, calms processing, evens dyeing and makes weight honest.
• Regain is reckoned on oven-dry mass; cotton's standard regain is 8.5 %; testing happens in the standard atmosphere, 20 °C / 65 % RH.
• Cellulosic yarns strengthen with moisture; wool and silk slightly weaken — the conditioning benefit list is fibre-dependent.
• Vacuum steam conditioning is the modern method because evacuation lets steam reach the package core evenly; room conditioning is slow and surface-first.
• Over-conditioning invites mildew, corrosion marks and sticking packages: target the standard, not the maximum.

Practice

1. A sample of yarn weighs 55.4 g and dries to 51.0 g. Calculate its regain and its moisture content, and state which figure a specification would use.
2. Weaving reports weft snarls and loops from a new yarn lot. Connect the fault to conditioning and state the corrective step in production.
3. Why does a plain steam autoclave still leave the core of a dense cone drier than its surface, while a vacuum cycle does not?
4. Give two distinct commercial (not processing) reasons a spinning mill conditions its yarn before dispatch.

Answer key

1. Regain = (55.4 − 51.0) ÷ 51.0 × 100 = 4.4 ÷ 51.0 × 100 ≈ **8.6 %**. Moisture content = 4.4 ÷ 55.4 × 100 ≈ **7.9 %**. Specifications and trade use **regain**.
2. Snarling is twist liveliness: the yarn was dispatched under-conditioned, so its twist is not set and it kinks whenever it goes slack in the weft path. Corrective step: condition the lot (vacuum steam conditioning for speed and evenness) to standard regain before weaving.
3. In a plain autoclave the package is full of air; steam condenses on the cool outer layers first, and the trapped air blocks penetration, so moisture advances slowly inward by diffusion and condensation. Evacuating first removes that air barrier: when steam is admitted it expands instantly through the whole package and condenses uniformly, core included.

4. (i) Weight: yarn is sold by weight at the agreed commercial regain — under-conditioned yarn is under-weight and the mill gives away or disputes money; (ii) count and quality claims: regain determines the measured count and test results, so conditioning to standard protects the mill's specifications and reputation (also acceptable: uniform dye uptake protecting shade claims).
